

Unified Detection of IT and Mobile Network Attacks

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Abstract: Network security is now more important due to the complexity of IT and mobile systems. The Edge-IIoTset dataset is used in this study to train a machine learning model that can recognize different attack types, including ransomware, port scanning, DDoS (UDP, ICMP, TCP), and bugin. The Python system is implemented using the Flask framework, which provides modules for user registration, login, dataset upload, model evaluation, prediction, and visualization. The effectiveness of algorithms such as Support Vector Machine, Random Forest, and XGBoost in classifying malicious activity is measured by accuracy, precision, recall, and F1-score. The simple interface used to display the results makes it easy for users to view predictions and model outcomes. By reducing manual monitoring effort and improving detection accuracy, the platform enables automated identification.

Key Words: Network attacks, IT security, mobile network, machine learning, dataset analysis, Flask, Python, prediction, model evaluation, attack detection

1. INTRODUCTION: Malicious attacks that jeopardize communications, data integrity, and system performance are a growing threat to IT and mobile networks in the current digital era. Secure network operations depend on the prompt and accurate identification of such threats. The Edge-IIoTset dataset is used in the "Detecting IT and Mobile Network Attacks" study to train machine learning models to recognize attacks like ransomware, port scanning, DDoS (UDP, ICMP, TCP), and Begin. To categorize traffic into normal or attack categories, algorithms such as Support Vector Machine, Random Forest, and XGBoost are used. Flask, which provides modules for dataset upload, model evaluation, prediction, and visualization, is used in the Python implementation of the system. This methodical approach improves automated network threat detection while reducing the need for manual monitoring.

2. LITERATURE REVIEW:

2.1 AILail, M., Garcia, A., & Olivo, S. (2023): This study uses random forest, SVM, logistic regression, and naive Bayes to compare machine learning models for intrusion detection. It draws attention to how class disparity and trait selection affect performance. According to the findings, ensemble techniques like random forests perform better than basic classifiers in terms of accuracy and detection rate. helpful in directing your project's model selection and preprocessing.

2.2 Talukdar, M.A., Islam, M.M., Uddin, M.A., et al. (2024): The study uses oversampling, stacking feature embedding, and PCA to address unbalanced, large-scale traffic datasets. In multi-class intrusion detection, these methods increase classification accuracy. The results

are useful for managing attack-type imbalances in the dataset for your project. offers doable strategies for boosting a search's dependability

2.3 Tseng, S.-M., Wang, Y.-Q., and Wang, Y.-C. (2024): The CIC IoT 2023 dataset (33 attack types) is used in this work to apply transformer models for IoT intrusion detection. The accuracy of the transformer was 99.40% when compared to DNN, CNN, RNN, and CNN+LSTM. demonstrates how deep learning can be used to detect multiple classes on big datasets. Motivate scaling strategies for detecting attacks on mobile and IT networks.

2.4 Alsaffar, A.M., Nouri Baygi, M., & Zolbanin, H.M. (2024): The authors suggest stacked ensemble learning (RF, CatBoost, XGBoost + MLP) and hybrid feature selection (MI Boruta). 99.92% accuracy for multi-class detection was attained on the CICIDS2017 dataset. shows how well feature selection and ensemble architecture work together. pertaining to creating reliable multi-algorithm detection systems for your project.

3. OBJECTIVES: This project's main goal is to create a machine learning system that can identify IT and mobile network attacks by categorizing traffic into groups like DDoS (UDP, ICMP, TCP), Port Scanning, Ransomware, and Begin. Uploading datasets, evaluating models, and creating forecasts are all made possible by an intuitive interface. The accuracy of algorithms such as SVM, Random Forest, and XGBoost is compared, and attack patterns are highlighted in the visualization. For strong network security, its modular architecture allows for scalability, new algorithms, and different kinds of attacks.

4. METHODOLOGY :

4.1 Random Forest: During training, the Random Forest method creates several independent decision trees as a collective learning technique. It effectively lowers the risk of overfitting associated with single decision trees by combining the predictions of these individual trees (voting for the final classification) to produce a highly stable and accurate result.

4.2 XGBoost (Extreme Gradient Boosting): This sophisticated gradient boosting framework builds collections of weak prediction models (decision trees) one after the other. It is very effective for managing complex feature interactions and attaining advanced performance because it concentrates on fixing the mistakes of earlier trees and uses gradient descent optimization to swiftly and effectively increase classification accuracy.

4.3 Support Vector Machine (SVM): The SVM technique divides data points into classes (such as normal vs. attack traffic) by identifying the best hyperplane in a high-dimensional feature space. It is robust and efficient for complicated classification tasks, particularly when working with high-dimensional feature sets, as it seeks to maximize the margin between the closest points of various classes (support vectors).

5.RESULTS :

5.1 Automated Detection Framework: The Edge-IIoTset dataset is used by the Python and Flask-built system to identify ransomware, port scanning, and DDoS attempts. It lowers human monitoring, offers dependable, scalable attack detection, and automates traffic analysis using machine learning models (SVM, Random Forest, XGBoost).

5.2 Algorithmic Outcomes and Performance: In all attack classes, ensemble techniques like Random Forest and XGBoost attained 100% accuracy, surpassing SVM's 96% accuracy with minor misclassification. These findings demonstrate that sophisticated machine learning algorithms are capable of removing false positives and offering reliable security monitoring for intricate networks.

6. ANALYSIS: The research effectively developed an automated approach for identifying IT and mobile network threats utilizing the Edge-IIoTset dataset based on system implementation and testing. After evaluating three machine learning models, Random Forest and XGBoost outperformed Support Vector Machine (SVM) models by a wide margin, achieving 96% accuracy despite minor misclassifications in some attack classes. The system successfully categorizes sophisticated threats like ransomware, port scanning, and DDoS (UDP, ICMP, TCP), confirming that ensemble approaches are best for managing high-dimensional network traffic data. Furthermore, operational viability was guaranteed by the incorporation of a Flask-based web interface, which enables users to upload data and obtain real-time, extremely precise forecasts with little manual interaction.

7. RECOMMENDATIONS: Future recommendations focus on transitioning systems from batch processing to real-time streaming analysis to enable immediate threat detection in live network traffic. To support the target environment, model optimization is needed for edge devices through techniques such as TinyML, which allows deployment on resource-constrained IoT nodes. Further work should expand the system's defensive capability by continuously updating it to classify a wider range of emerging zero-day and sophisticated attacks such as APT. Finally, the operational value of the system must be increased by implementing an attribute importance visualization module for clarity and developing a mechanism for automated response actions to immediately mitigate confirmed threats.

8. LIMITATIONS: The current system's operational environment is the main source of its limitations. Specifically, detection is delayed since analysis is done in batch processing of pre-aggregated datasets rather than in real-time. Furthermore, the Edge-IIoTset dataset's scope limits the system's efficacy because it hasn't been tested against zero-days or more complex, new attack vectors that aren't included in the training data. Lastly, the developed web application is a proof-of-concept interface and does not have the characteristics needed for a reliable, enterprise-level deployment, including automated response mechanisms, comprehensive logging, or scalability for high network traffic levels.

9. CONCLUSION: The project successfully developed a system for "Detection of IT and Mobile Network Attacks" using SVM, Random Forest, and XGBoost machine learning models. The implemented system handles the entire process, from dataset loading and preprocessing to training and prediction, and includes a secure, user-friendly interface. Extensive testing confirmed the system is reliable, accurate, and stable, achieving high performance metrics across various attack categories. Overall, the project provides a robust, modular framework that enhances network security by effectively monitoring, classifying, and alerting users to malicious activities.

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